

1. The Z Transform

- Powerful tool for analyzing & designing DT systems
- Generalization of the DTFT:

$$G(z) = \mathcal{Z}\{g[n]\} = \sum_{n=-\infty}^{\infty} g[n]z^{-n}$$
Z Transform
- z is complex...
 - $z = e^{j\omega} \rightarrow$ DTFT
 - $z = re^{j\omega} \rightarrow \sum_n g[n]r^{-n}e^{-j\omega n}$ **DTFT of $r^n \cdot g[n]$**

ROC is necessary!

- To completely define a ZT, **you must specify the ROC**:

$$x[n] = \lambda^n u[n] \rightarrow X(z) = \frac{1}{1 - \lambda z^{-1}}$$
ROC $|z| > |\lambda|$
- $$x[n] = -\lambda^n u[-n-1] \rightarrow X(z) = \frac{1}{1 - \lambda z^{-1}}$$
ROC $|z| < |\lambda|$
- A single $G(z)$ can describe several sequences with different ROCs

Rational Z transforms

- $G(z)$ can be any function; **rational polynomials** are important class:

$$G(z) = \frac{P(z)}{D(z)} = \frac{p_0 + p_1 z^{-1} + \dots + p_{M-1} z^{-(M-1)} + p_M z^{-M}}{d_0 + d_1 z^{-1} + \dots + d_{N-1} z^{-(N-1)} + d_N z^{-N}}$$
- $$\sum_{k=0}^{\infty} (ar^k) = a \left(\frac{1-r^{N+1}}{1-r} \right)$$

Factored rational ZTs

- Numerator, denominator can be **factored**:

$$G(z) = \frac{p_0 \prod_{l=1}^M (1 - \zeta_l z^{-1})}{d_0 \prod_{l=1}^N (1 - \lambda_l z^{-1})} = \frac{z^M p_0 \prod_{l=1}^M (z - \zeta_l)}{z^N d_0 \prod_{l=1}^N (z - \lambda_l)}$$
- $\{\zeta_l\}$ are roots of numerator $\rightarrow G(z) = 0 \rightarrow \{\zeta_l\}$ are the **zeros** of $G(z)$
- $\{\lambda_l\}$ are roots of denominator $\rightarrow G(z) = \infty \rightarrow \{\lambda_l\}$ are the **poles** of $G(z)$

ROCs and sidedness

- LCCDEs have solutions of form:

$$y_c[n] = \alpha_i \lambda_i^n u[n] + \dots$$
- Hence ZT with terms like $\frac{\alpha_i}{1 - \lambda_i z^{-1}} + \dots$
- Each **pole** λ_i of $G(z)$ corresponds to a **term** λ_i^n in $g[n]$
- We saw for $G(z) = \frac{1}{1 - \lambda z^{-1}}$:
 - ROC $|z| > |\lambda| \rightarrow g[n] = \lambda^n u[n]$ **RIGHT-SIDED**
 - ROC $|z| < |\lambda| \rightarrow g[n] = -\lambda^n u[-n-1]$ **LEFT-SIDED**

Some common Z transforms

$g[n]$	$G(z)$	ROC
$\delta[n]$	1	$\forall z$
$u[n]$	$\frac{1}{1 - z^{-1}}$	$ z > 1$
$\alpha^n u[n]$	$\frac{1}{1 - \alpha z^{-1}}$	$ z > \alpha $
$r^n \cos(\omega_0 n) u[n]$	$\frac{1 - r \cos(\omega_0) z^{-1}}{1 - 2r \cos(\omega_0) z^{-1} + r^2 z^{-2}}$	$ z > r$ sum of $r^n e^{j\omega_0 n} + r^n e^{-j\omega_0 n}$
$r^n \sin(\omega_0 n) u[n]$	$\frac{r \sin(\omega_0) z^{-1}}{1 - 2r \cos(\omega_0) z^{-1} + r^2 z^{-2}}$	$ z > r$ "conjugate pole pair"

Z Transform properties

	$g[n] \leftrightarrow G(z)$	w/ROC $\mathcal{R}_g$
Conjugation	$g^*[n] \leftrightarrow G^*(z^*)$	$\mathcal{R}_g$
Time reversal	$g[-n] \leftrightarrow G(1/z)$	$1/\mathcal{R}_g$
Time shift	$g[n - n_0] \leftrightarrow z^{-n_0} G(z)$	$\mathcal{R}_g$ ($0/\infty$?)
Exp. scaling	$\alpha^n g[n] \leftrightarrow G(z/\alpha)$	$ \alpha \mathcal{R}_g$
Diff. wrt z	$ng[n] \leftrightarrow -z \frac{dG(z)}{dz}$	$\mathcal{R}_g$ ($0/\infty$?)

Z Transform properties

	$g[n] \leftrightarrow G(z)$	ROC
Convolution	$g[n] \otimes h[n] \leftrightarrow G(z)H(z)$	at least $\mathcal{R}_g \cup \mathcal{R}_h$
Modulation	$g[n]h[n] \leftrightarrow \frac{1}{2\pi j} \oint_{\mathcal{C}} G(v)H\left(\frac{z}{v}\right)v^{-1}dv$	at least $\mathcal{R}_g \cap \mathcal{R}_h$

ZT Example

- $x[n] = r^n \cos(\omega_0 n) u[n]$; can express as

$$\frac{1}{2} u[n] \left((re^{j\omega_0})^n + (re^{-j\omega_0})^n \right) = v[n] + v^*[n]$$
- Hence, $X(z) = V(z) + V^*(z^*)$

$$= \frac{1}{2} \left(\frac{1}{1 - re^{j\omega_0} z^{-1}} + \frac{1}{1 - re^{-j\omega_0} z^{-1}} \right)$$

$$= \frac{1 - r \cos(\omega_0) z^{-1}}{1 - 2r \cos(\omega_0) z^{-1} + r^2 z^{-2}}$$

IZT#3: Partial Fractions

- Basic idea: Rearrange $G(z)$ as **sum** of terms *recognized* as simple ZTs
 - especially $\frac{1}{1 - \alpha z^{-1}} \leftrightarrow \alpha^n u[n]$ or sin/cos forms
- i.e. given products

$$\frac{P(z)}{(1 - \alpha z^{-1})(1 - \beta z^{-1}) \dots}$$
 rearrange to sums

$$\frac{A}{1 - \alpha z^{-1}} + \frac{B}{1 - \beta z^{-1}} + \dots$$

- Procedure:

$$F(z) = \frac{P(z)}{\prod_{l=1}^N (1 - \lambda_l z^{-1})} = \sum_{l=1}^N \frac{\rho_l}{1 - \lambda_l z^{-1}}$$

$$\rightarrow f[n] = \sum_{l=1}^N \rho_l (\lambda_l)^n u[n]$$
- where $\rho_l = (1 - \lambda_l z^{-1}) F(z) \Big|_{z=\lambda_l}$ (cancels term in denominator) i.e. evaluate $F(z)$ at the pole but **multiplied** by the pole term

IZT #2: Long division

- e.g. $H(z) = \frac{1 + 2z^{-1}}{1 + 0.4z^{-1} - 0.12z^{-2}}$

$$= \frac{1 + 1.6z^{-1} - 0.52z^{-2} + 0.4z^{-3}}{1 + 0.4z^{-1} - 0.12z^{-2}}$$

$$= \frac{1.6z^{-1} + 0.12z^{-2}}{1.6z^{-1} + 0.64z^{-2} - 0.192z^{-3}} + \frac{-0.52z^{-2} + 0.192z^{-3}}{1.6z^{-1} + 0.64z^{-2} - 0.192z^{-3}}$$

IZT #1: Generalize IDTFT

- If $z = re^{j\omega}$ then

$$G(z) = G(re^{j\omega}) = \sum g[n] r^{-n} e^{-j\omega n} = \text{DTFT}\{g[n]r^{-n}\}$$
- so $g[n]r^{-n} = \frac{1}{2\pi} \int_{-\pi}^{\pi} G(re^{j\omega}) e^{j\omega n} d\omega$ **IDTFT**

$$= \frac{1}{2\pi j} \oint_{\mathcal{C}} G(z) z^{n-1} dz$$
- Any closed contour around origin will do
- DTFT example
 - e.g. $x[n] = \alpha^n u[n]$, $|\alpha| < 1$

$$\Rightarrow X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} \alpha^n u[n] e^{-j\omega n}$$

ROC intersections

- 2 poles, $|\alpha| < |\beta|$, give several possible ROCs:
 - $\alpha^n u[n] + \beta^n u[n]$ both **right-sided** ROC $|z| > |\alpha|, |z| > |\beta|$
 - $\alpha^n u[-n] + \beta^n u[-n]$ both **left-sided** ROC $|z| < |\alpha|, |z| < |\beta|$
 - $\alpha^n u[n] + \beta^n u[-n]$ **two-sided** ROC $|\alpha| < |z| < |\beta|$
 - $\alpha^n u[-n] + \beta^n u[n]$ **two-sided** ROC $|z| < |\alpha|, |z| > |\beta|$
- no ROC means...
- Note: **Two-sided exponential**

$$g[n] = \alpha^n \quad -\infty < n < \infty$$

$$= \alpha^n u[n] + \alpha^n u[-n-1]$$

- No overlap in ROCs $\rightarrow$ ZT does not exist

1. The Fourier Transform

- Basic observation (continuous time): A periodic signal can be decomposed into sinusoids at **integer multiples** of the **fundamental frequency**
- i.e. if $\tilde{x}(t) = \tilde{x}(t + T)$ we can approach $\tilde{x}$ with

$$\tilde{x}(t) \approx \sum_{k=0}^M a_k \cos\left(\frac{2\pi k}{T} t + \phi_k\right)$$

Fourier domain

- x is equivalently described by its Fourier Series parameters:

$$a_k = (-1)^{\frac{k-1}{2}} \frac{1}{k} \quad k = 1, 3, 5, \dots$$
- Complex form: $\tilde{x}(t) \approx \sum_{k=-M}^M c_k e^{j\frac{2\pi k}{T} t}$
- How to find $\{c_k\}$, $\{\arg\{c_k\}\}$? **Inner product** with complex sinusoids:

$$c_k = \frac{1}{2\pi} \int_{-T/2}^{T/2} x(t) e^{-j\frac{2\pi k}{T} t} dt$$

$$= \frac{1}{2\pi} \left(\int x(t) \cos \frac{j2\pi k t}{T} dt - j \int x(t) \sin \frac{j2\pi k t}{T} dt \right)$$

sinc

- $\text{sinc } x = \frac{\sin x}{x}$
- $= 1$ when $x = 0$
- $= 0$ when $x = r\pi$, $r \neq 0$, $r = \pm 1, \pm 2, \pm 3, \dots$

Fourier Transform

- Fourier series for periodic signals extends naturally to **Fourier Transform** for any (CT) signal (not just periodic):

$$X(\Omega) = \frac{1}{2\pi} \int_{-\infty}^{\infty} x(t) e^{-j\Omega t} dt$$
 Fourier Transform (FT)
- $$x(t) = \int_{-\infty}^{\infty} X(\Omega) e^{j\Omega t} d\Omega$$
 Inverse Fourier Transform (IFT)
- Discrete index $k \rightarrow$ continuous freq. Ω

(continuous time) i.e.

$\int \delta(x - x_0) f(x) dx = f(x_0)$

$\rightarrow x(t) = A e^{j\Omega_0 t} \leftrightarrow X(\Omega) = A \delta(\Omega - \Omega_0)$

Fourier Transform of a sine

Assume $x(t) = e^{j\Omega_0 t}$

Now, since $x(t) = \int_{-\infty}^{\infty} X(\Omega) e^{j\Omega t} d\Omega$

we know $X(\Omega) = \delta(\Omega - \Omega_0)$

where $\delta(x)$ is the Dirac delta function

2. Discrete Time FT (DTFT)

FT defined for discrete sequences:

$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$ **DTFT**

Periodicity of $X(e^{j\omega})$

- $X(e^{j\omega})$ has periodicity 2π in ω :

$$X(e^{j(\omega+2\pi)}) = \sum x[n]e^{-j(\omega+2\pi)n} = \sum x[n]e^{-j\omega n}e^{-j2\pi n} = X(e^{j\omega})$$

Inverse DTFT (IDTFT)

- Same basic form as other IFTs:

$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega})e^{j\omega n} d\omega \quad \text{IDTFT}$$

- Note: continuous, periodic $X(e^{j\omega})$
discrete, infinite $x[n]$...

$$g[n] \otimes h[n] \leftrightarrow G(e^{j\omega})H(e^{j\omega}) \quad \text{Convolution becomes multiplication}$$

Energy density spectrum

- Energy of sequence $\varepsilon_g = \sum_{\forall n} |g[n]|^2$
- By Parseval $\varepsilon_g = \frac{1}{2\pi} \int_{-\pi}^{\pi} |G(e^{j\omega})|^2 d\omega$
- Define **Energy Density Spectrum (EDS)**

$$S_{gg}(e^{j\omega}) = |G(e^{j\omega})|^2$$

System Response from $H(e^{j\omega})$

- If $x[n]$ is a **cpx sinusoid** at ω_0
then the output of a system with **IR** $h[n]$ is the **same sinusoid** scaled by $|H(e^{j\omega_0})|$ and phase-shifted by $\arg\{H(e^{j\omega_0})\} = \theta(\omega)$ where $H(e^{j\omega}) = \text{DTFT}\{h[n]\}$

(Any signal can be expressed as sines...)

- $|H(e^{j\omega})|$ "magnitude response" $\rightarrow$ gain
- $\arg\{H(e^{j\omega})\}$ "phase resp." $\rightarrow$ phase shift

TF $\rightarrow$ FR

- DTFT $H(e^{j\omega}) = \text{ZT } H(z)|_{z=e^{j\omega}}$
i.e. **Frequency Response** is **Transfer Function** eval'd on **Unit Circle**
- factor:
- $$H(z) = \frac{p_0 \prod_{k=1}^M (1 - \zeta_k z^{-1})}{d_0 \prod_{k=1}^N (1 - \lambda_k z^{-1})} = \frac{p_0 z^{-M} \prod_{k=1}^M (z - \zeta_k)}{d_0 z^{-N} \prod_{k=1}^N (z - \lambda_k)}$$
- $$\Rightarrow H(e^{j\omega}) = \frac{p_0}{d_0} e^{j\omega(N-M)} \frac{\prod_{k=1}^M (e^{j\omega} - \zeta_k)}{\prod_{k=1}^N (e^{j\omega} - \lambda_k)}$$

DTFT of simple sequences

- $x[n] = e^{j\omega_0 n}$: $x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega})e^{j\omega n} d\omega$ IDTFT
 $\Rightarrow X(e^{j\omega}) = 2\pi \cdot \delta(\omega - \omega_0)$ over $-\pi < \omega < \pi$
but $X(e^{j\omega})$ must be **periodic** in $\omega \Rightarrow$

$$e^{j\omega_0 n} \leftrightarrow \sum_k 2\pi \cdot \delta(\omega - \omega_0 - 2\pi k)$$
- If $\omega_0 = 0$ then $x[n] = 1 \forall n$
so $1 \leftrightarrow \sum_k 2\pi \cdot \delta(\omega - 2\pi k)$

DTFT of real $x[n]$

- When $x[n]$ is pure real, $\Rightarrow X(e^{j\omega}) = X^*(e^{-j\omega})$
- $x_{cs}[n] \equiv x_{ev}[n] = x_{ev}[-n] \Leftrightarrow X_R(e^{j\omega}) = X_R(e^{-j\omega})$
- $x_{od}[n] \equiv x_{od}[n] = -x_{od}[-n] \Leftrightarrow X_I(e^{j\omega}) = -X_I(e^{-j\omega})$

EDS and autocorrelation

- Autocorrelation of $g[n]$:

$$r_{gg}[\ell] = \sum_{n=-\infty}^{\infty} g[n]g[n-\ell] = g[n] \otimes g[-n]$$

$$\Rightarrow \text{DTFT}\{r_{gg}[\ell]\} = G(e^{j\omega})G(e^{-j\omega})$$
- If $g[n]$ is **real**, $G(e^{j\omega}) = G^*(e^{-j\omega})$, so

$$\text{DTFT}\{r_{gg}[\ell]\} = |G(e^{j\omega})|^2 = S_{gg}(e^{j\omega})$$
 no phase info.
- Mag-sq of spectrum is DTFT of autocorrelation

Transient / Steady State

- Most signals start at a finite time e.g.
 $x[n] = e^{j\omega_0 n} u[n]$ What is the effect?
- $y[n] = h[n] \otimes x[n] = \sum_{k=-\infty}^n h[k]e^{j\omega_0(n-k)} = \sum_{k=-\infty}^n h[k]e^{j\omega_0 n}e^{-j\omega_0 k} = H(e^{j\omega_0})e^{j\omega_0 n} - \left(\sum_{k=n}^{\infty} h[k]e^{-j\omega_0 k}\right)e^{j\omega_0 n}$

Steady state same as with pure sine input
Transient response - consequence of gating

TF $\rightarrow$ FR

- $H(e^{j\omega}) = \frac{p_0}{d_0} e^{j\omega(N-M)} \frac{\prod_{k=1}^M (e^{j\omega} - \zeta_k)}{\prod_{k=1}^N (e^{j\omega} - \lambda_k)}$ ζ_k, λ_k are TF roots on z-plane
- $\Rightarrow |H(e^{j\omega})| = \left| \frac{p_0}{d_0} \right| \frac{\prod_{k=1}^M |e^{j\omega} - \zeta_k|}{\prod_{k=1}^N |e^{j\omega} - \lambda_k|}$ **Magnitude response**
- $\theta(\omega) = \arg\left\{\frac{p_0}{d_0}\right\} + \omega(N-M) + \sum_{k=1}^M \arg\{e^{j\omega} - \zeta_k\} - \sum_{k=1}^N \arg\{e^{j\omega} - \lambda_k\}$ **Phase response**

DTFT properties

- Linear:
 $\alpha g[n] + \beta h[n] \Leftrightarrow \alpha G(e^{j\omega}) + \beta H(e^{j\omega})$
- Time shift:
 $g[n - n_0] \Leftrightarrow e^{-j\omega n_0} G(e^{j\omega})$
- Frequency shift:
 $e^{j\omega_0 n} g[n] \Leftrightarrow G(e^{j(\omega - \omega_0)})$ 'delay' in frequency

DTFT modulation

- Modulation: $x[n] = g[n] \cdot h[n]$
Could solve if $g[n]$ was just sinusoids...

$$g[n] \cdot h[n] \Leftrightarrow \frac{1}{2\pi} \int_{-\pi}^{\pi} G(e^{j\theta})H(e^{j(\omega - \theta)})d\theta$$
 Dual of convolution in time

Convolution with DTFT

- Since $g[n] \otimes h[n] \Leftrightarrow G(e^{j\omega})H(e^{j\omega})$
we can calculate a convolution by:
 - finding DTFTs of $g, h \rightarrow G, H$
 - multiply them: $G \cdot H$
 - IDTFT of product is result, $g \otimes h$

2. Transfer Function (TF)

Linking LCCDE, ZT & Freq. Resp...

- LCCDE: $\sum_{k=0}^N d_k y[n-k] = \sum_{k=0}^M p_k x[n-k]$
- Take ZT: $\sum_{k=0}^N d_k z^{-k} Y(z) = \sum_{k=0}^M p_k z^{-k} X(z)$
- Hence: $Y(z) = \frac{\sum_{k=0}^M p_k z^{-k}}{\sum_{k=0}^N d_k z^{-k}} X(z)$ **Transfer function** $H(z)$

or:

DTFT symmetry

- If $x[n] \Leftrightarrow X(e^{j\omega})$ then...
 $x[-n] \Leftrightarrow X(e^{-j\omega})$ **from summation**
 $x^*[n] \Leftrightarrow X^*(e^{-j\omega})$ $(e^{-j\omega})^* = e^{j\omega}$
 $\text{Re}\{x[n]\} \Leftrightarrow X_{CS}(e^{j\omega}) = \frac{1}{2}[X(e^{j\omega}) + X^*(e^{-j\omega})]$ **conjugate symmetry cancels Im parts on IDTFT**
 $j\text{Im}\{x[n]\} \Leftrightarrow X_{CA}(e^{j\omega}) = \frac{1}{2j}[X(e^{j\omega}) - X^*(e^{-j\omega})]$

Parseval's relation

- "Energy" in time and frequency domains are equal:

$$\sum_{\forall n} g[n]h^*[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} G(e^{j\omega})H^*(e^{j\omega})d\omega$$
- If $g = h$, then $g \cdot g^* = |g|^2 = \text{energy}$...

TABLE 3.2 SOME z-TRANSFORM PROPERTIES

Section Reference	Sequence	Transform	ROC
	$x[n]$	$X(z)$	R_x
	$x_1[n]$	$X_1(z)$	R_{x_1}
	$x_2[n]$	$X_2(z)$	R_{x_2}
3.4.1	$ax_1[n] + bx_2[n]$	$aX_1(z) + bX_2(z)$	Contains $R_{x_1} \cap R_{x_2}$
3.4.2	$x[n - n_0]$	$z^{-n_0} X(z)$	R_x , except for the possible addition or deletion of the origin or ∞
3.4.3	$z_0^n x[n]$	$X(z/z_0)$	$ z_0 R_x$
3.4.4	$nx[n]$	$-z \frac{dX(z)}{dz}$	R_x , except for the possible addition or deletion of the origin or ∞
3.4.5	$x^*[n]$	$X^*(z^*)$	R_x
	$\text{Re}\{x[n]\}$	$\frac{1}{2}[X(z) + X^*(z^*)]$	Contains R_x
	$\text{Im}\{x[n]\}$	$\frac{1}{2j}[X(z) - X^*(z^*)]$	Contains R_x
3.4.6	$x^*[-n]$	$X^*(1/z^*)$	$1/R_x$
3.4.7	$x_1[n] * x_2[n]$	$X_1(z)X_2(z)$	Contains $R_{x_1} \cap R_{x_2}$
3.4.8	Initial-value theorem: $x[n] = 0, \quad n < 0 \quad \lim_{z \rightarrow \infty} X(z) = x[0]$		